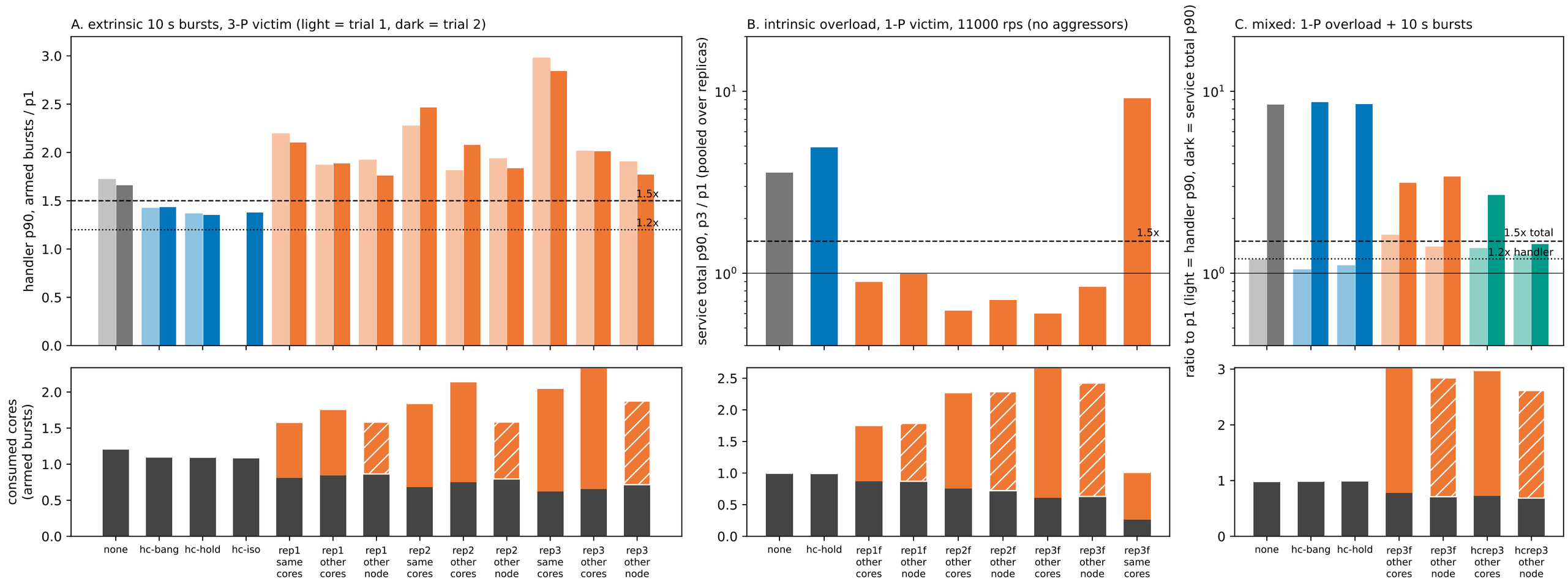
Motivation pass 2 (sm110p, July layout): latency vs consumed cores per mitigation and placement
A. extrinsic 10 s bursts, 3-P victim (light = trial 1, dark = trial 2)
handler p90, armed bursts / p1
1.5x
1.2x
B. intrinsic overload, 1-P victim, 11000 rps (no aggressors)
service total p90, p3 / p1 (pooled over replicas)
1.5x
C. mixed: 1-P overload + 10 s bursts
ratio to p1 (light = handler p90, dark = service total p90)
1.5x total
1.2x handler
consumed cores (armed bursts)
none
hc-bang
hc-hold
hc-iso
rep1 same cores
rep1 other cores
rep1 other node
rep2 same cores
rep2 other cores
rep2 other node
rep3 same cores
rep3 other cores
rep3 other node
rep1f other cores
rep1f other node
rep2f other cores
rep2f other node
rep3f other cores
rep3f other node
rep3f same cores
hcrep3 other cores
hcrep3 other node
no mitigation
harvesting (HC)
replication
HC + replicas
cores: victim
cores: replicas on node-4
cores: replicas on node-3